

TOHOLAMPI LAITALA, Finland

WMO: 027370

Lat: 63.8214N

Lon: 24.1633E

Elev: 277

StdP: 14.55

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.0	-10.6	-21.2	1.7	-16.3	-14.3	2.6	-10.0	24.8	28.9	22.5	27.2	3.2	170	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	79.6	65.4	75.7	62.8	72.6	61.4	67.4	76.0	65.2	72.9	63.2	70.2	8.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.6	92.0	70.8	62.3	84.8	68.3	60.2	78.7	66.9	32.0	76.2	30.2	73.0	28.8	70.0	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	18.6	16.6	DB	-21.1	83.3	6.5	4.3	-25.8	86.4	-29.5	88.9	-33.2	91.3	-37.9	94.4
			WB	-20.7	70.3	6.4	3.5	-25.4	72.8	-29.1	74.9	-32.8	76.8	-37.4	79.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	15.4	18.3	25.0	35.9	47.8	54.7	60.5	57.3	49.5	37.5	31.2	23.9
	DBStd	17.52	13.41	14.23	9.67	6.66	8.33	6.89	6.16	5.73	6.05	6.92	8.36	11.37
	HDD50	5184	1072	889	774	423	142	28	3	8	81	389	565	810
	HDD65	9832	1537	1309	1239	872	535	317	168	248	465	852	1015	1275
	CDD50	872	0	0	0	1	73	169	327	234	66	2	0	0
	CDD65	46	0	0	0	0	2	8	27	9	0	0	0	0
	CDH74	511	0	0	0	0	44	94	295	78	0	0	0	0
	CDH80	77	0	0	0	0	1	12	52	12	0	0	0	0
Wind	WSAvg	8.2	8.4	9.0	9.0	7.7	8.4	8.3	6.6	7.1	7.6	8.0	8.6	9.5
Precipitation	PrecAvg	23.7	1.6	1.3	1.2	1.3	1.8	2.5	3.1	2.8	2.1	2.2	2.0	1.9
	PrecMax	30.4	3.0	2.5	2.0	2.8	3.4	4.5	5.4	5.6	4.9	4.2	3.8	3.9
	PrecMin	18.0	0.4	0.2	0.2	0.3	0.3	0.8	0.8	1.1	0.7	0.5	0.2	0.7
	PrecStd	3.2	0.7	0.6	0.5	0.6	0.9	1.1	1.1	1.2	1.2	1.0	0.8	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	42.1	46.5	64.9	78.6	81.8	84.7	81.9	68.4	56.8	49.9	41.9
		MCWB	36.9	38.2	38.6	49.1	59.6	65.0	66.8	66.8	59.9	53.7	47.6	40.4
	2%	DB	35.9	39.4	43.0	56.2	74.6	76.6	81.0	75.3	65.1	52.3	46.0	39.9
		MCWB	34.7	36.1	37.4	45.2	58.2	64.0	67.2	63.1	58.0	50.0	44.2	38.4
	5%	DB	33.9	36.5	39.5	50.9	70.0	71.9	77.0	71.6	61.9	50.0	43.1	37.6
		MCWB	33.1	34.8	35.5	43.3	56.5	59.8	64.8	61.4	55.5	48.0	42.2	36.2
	10%	DB	32.5	34.3	37.0	47.2	64.1	68.2	73.6	68.2	59.4	47.5	40.9	35.4
		MCWB	31.8	33.2	33.9	40.7	53.1	58.5	63.5	60.0	54.7	45.5	40.1	34.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	38.9	40.7	50.1	62.1	67.3	71.5	69.9	61.1	53.9	48.1	41.0
		MCDB	38.1	41.9	45.2	58.9	73.4	77.7	80.3	77.7	66.3	55.6	49.6	41.2
	2%	WB	35.1	36.7	38.0	47.2	59.6	65.1	68.6	65.4	59.0	50.7	44.6	38.6
		MCDB	35.8	38.8	41.3	55.8	71.6	74.3	77.0	71.9	63.7	52.0	45.7	39.5
	5%	WB	33.3	35.0	36.1	43.8	57.4	62.5	66.4	63.3	57.0	48.3	42.2	36.3
		MCDB	33.8	36.6	39.0	49.5	68.6	70.6	74.4	68.8	60.3	49.8	43.0	37.2
	10%	WB	31.9	33.3	34.3	41.3	54.4	59.8	64.4	61.5	55.5	45.7	40.2	34.5
		MCDB	32.6	34.1	36.3	46.1	63.9	66.6	72.0	66.8	58.8	47.3	40.8	35.2
Mean Daily Temperature Range	5% DB	MDBR	11.2	12.3	15.1	15.6	18.9	17.7	18.0	17.3	14.2	10.2	7.0	8.6
		MCDBR	7.2	9.1	13.6	24.4	28.9	24.7	24.3	23.1	19.0	12.0	8.1	7.8
		MCWBR	7.0	7.7	9.8	15.4	14.9	13.1	12.2	12.0	12.2	9.9	7.5	7.6
	5% WB	MCDBR	7.3	8.7	12.3	21.8	27.3	21.2	21.6	18.4	16.2	11.3	8.0	8.1
		MCWBR	7.2	7.6	9.4	15.0	14.7	11.8	11.6	10.6	11.1	9.8	7.5	8.0
Clear-Sky Solar Irradiance	taub	0.221	0.283	0.311	0.350	0.347	0.348	0.369	0.353	0.330	0.309	0.363	0.354	
	taud	2.003	2.146	2.212	2.312	2.444	2.466	2.427	2.464	2.515	2.378	2.639	2.612	
	Ebn at Noon	115	199	245	262	275	277	266	260	244	193	106	58	
	Edh at Noon	9	20	28	32	31	31	31	27	21	16	8	5	
All-Sky Solar Radiation	RadAvg	47	224	698	1195	1576	1735	1608	1219	694	261	62	19	
	RadStd	5	42	70	114	168	150	157	122	74	48	11	2	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page